BT612T – FERMENTATION TECHNOLOGY

FINAL TERM IMPORTANT QUESTIONS

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Admin: **BS Biological Sciences** (One place to all your solutions)

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Short Questions

1. What is positive and negative selection?

Positive Selection

Positive selection is a method used to isolate desired microorganisms or mutants in which only the organisms possessing a specific trait are able to grow on a selective medium, while all other organisms fail to grow. For example, when bacteria are grown on a medium containing ampicillin, only ampicillin-resistant bacteria survive and form colonies, while sensitive bacteria die.

Negative Selection

Negative selection is a method used to isolate desired mutants by eliminating or inhibiting the growth of undesired or wild-type cells, allowing the required mutants to be identified from the surviving population. For example, in the penicillin enrichment technique, penicillin kills actively growing wild-type bacteria, whereas auxotrophic mutants, which are not growing due to nutrient deficiency, survive and can be isolated later.

2. Write Baltimore method of oxygen transfer.

Baltimore Method of Oxygen Transfer

The Baltimore method explains the transfer of oxygen from air to microbial cells during aerobic fermentation. It occurs in the following steps:

- ***Transfer from air bubble to liquid surface:*** Oxygen moves from the air bubble to the gas–liquid interface.
- ***Diffusion through the liquid boundary layer:*** Oxygen diffuses across the thin liquid film surrounding the air bubble. This is usually the rate-limiting step in oxygen transfer.
- ***Movement into the bulk liquid:*** Oxygen dissolves in the fermentation broth and is distributed throughout the medium by agitation.
- ***Uptake by microbial cells:*** Dissolved oxygen diffuses to the surface of microbial cells and is utilized for respiration, growth, and product formation.

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3. What are trace elements?

Trace elements are inorganic mineral elements that are required by microorganisms in very small (trace) amounts for normal growth and metabolism. They mainly act as cofactors or activators of enzymes, helping in various biochemical and metabolic reactions. Although needed in minute quantities, their absence can reduce microbial growth and fermentation efficiency. Common trace elements used in fermentation media include iron (Fe), zinc (Zn), manganese (Mn), copper (Cu), cobalt (Co), molybdenum (Mo), and nickel (Ni).

4. How oxygen is supplied to microbial culture during fermentation?

Oxygen is supplied to microbial cultures during fermentation by aeration and agitation. In aeration, sterile air containing oxygen is introduced into the fermenter through a sparger, which breaks air into small bubbles and increases the surface area for oxygen transfer. Agitation by mechanical impellers helps in uniform mixing of the culture medium, disperses air bubbles, and improves oxygen availability to microbial cells. The efficiency of oxygen supply depends on factors such as air flow rate, agitation speed, temperature, pressure, and oxygen transfer coefficient (KLa).

5. Write name of some antifoams.

Some commonly used antifoams in fermentation are **si**licone oils (polydimethylsiloxane), vegetable oils, mineral oils, polypropylene glycol, lard oil, octanol, and oleic acid**.** These compounds reduce foam formation by decreasing surface tension and destabilizing foam bubbles in the fermentation medium.

6. Enlist any five inhibitors that are used in fermentation.

Some inhibitors used in fermentation are:

1. ***Penicillin*** – inhibits growth of actively dividing bacteria.
2. ***Sodium azide*** – inhibits respiratory enzymes and microbial growth.
3. ***Chloroform*** – used to inhibit unwanted microbial contamination.
4. ***Phenol*** – acts as an antimicrobial agent.
5. ***Sodium benzoate*** – inhibits growth of unwanted microorganisms.

7. Write factors affecting KLa.

The factors affecting KLa (volumetric oxygen transfer coefficient) are:

- ***Agitation speed*** – Higher agitation increases bubble dispersion and improves oxygen transfer.
- ***Aeration rate*** – Increasing the air flow rate increases oxygen availability in the fermenter.
- ***Impeller design and size*** – Impeller type, diameter, and number affect mixing and oxygen transfer efficiency.
- ***Viscosity of fermentation broth*** – High viscosity reduces oxygen diffusion and lowers KLa.
- ***Temperature and pressure*** – These influence oxygen solubility and transfer rate.
- ***Presence of antifoams*** – Antifoaming agents may decrease oxygen transfer by reducing bubble stability.
- ***Size of air bubbles*** – Smaller bubbles provide greater surface area and increase oxygen transfer.

8. What is lyophilization? What are its advantages?

Lyophilization, also known as freeze drying, is a method of preservation in which microorganisms are frozen at low temperatures and the water content is removed by sublimation under vacuum. It helps maintain the viability and stability of microbial cultures for long-term storage.

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Advantages:

- Maintains the viability and genetic stability of microorganisms.
- Allows long-term preservation of cultures.
- Reduces metabolic activity and prevents microbial deterioration.
- Cultures can be easily transported and stored.
- Dried cultures can be rapidly revived by rehydration.

9. Write three ways to reduce foam.

Foam formation during fermentation can be reduced by the following methods:

- ***Addition of antifoaming agents:*** Chemical antifoams such as silicone oils, vegetable oils, and polypropylene glycol are added to the fermentation medium. They reduce surface tension and prevent the formation and stability of foam bubbles.
- ***Mechanical foam breaking:*** Foam can be controlled by using mechanical foam breakers or special rotating devices that physically collapse foam bubbles during fermentation.
- ***Control of aeration and agitation:*** Excessive air supply and high agitation speed increase foam formation. Maintaining optimum aeration and agitation conditions helps reduce excessive foam production in the fermenter.

10. Write Johnson's equation.

Johnson's equation is used to describe the relationship between oxygen transfer rate (OTR) and oxygen uptake rate (OUR) during fermentation.

$$OTR = K_L a\ (C^* - C_L)$$

Where:

- OTR = Oxygen transfer rate (mg/L/h)
- KLa = Volumetric oxygen transfer coefficient
- C* = Saturation concentration of oxygen in the liquid
- C_l = Actual dissolved oxygen concentration in the fermentation broth

This equation shows that oxygen transfer increases with an increase in KLa and the difference between saturation and actual dissolved oxygen concentration.

11. Write limitations of using starch in fermentation medium.

The limitations of using starch as a carbon source in fermentation medium are:

- ***Low solubility:*** Starch has poor solubility in water, making it difficult to mix uniformly in the fermentation medium.
- ***Need for hydrolysis:*** Microorganisms cannot directly utilize complex starch; it must be broken down into simpler sugars by enzymes or acid treatment.
- ***High viscosity:*** Starch increases the viscosity of the medium, which reduces mixing and oxygen transfer efficiency.
- ***Sterilization problems:*** High starch concentration may cause difficulties during sterilization due to gelatinization and clumping.
- ***Incomplete utilization:*** Some microorganisms lack the enzymes required to degrade starch efficiently, resulting in lower product yield.

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12. Differentiate between aerobic and anaerobic fermentation.

Aerobic fermentation	Anaerobic fermentation
Requires the presence of oxygen for microbial growth and product formation.	Occurs in the absence of oxygen.
Oxygen is supplied through aeration and agitation in the fermenter.	No external oxygen supply is required.
Microorganisms use oxygen as the final electron acceptor during respiration.	Microorganisms use organic compounds or other molecules as electron acceptors.
Produces more energy due to complete oxidation of substrates.	Produces less energy due to incomplete oxidation of substrates.
Examples include production of penicillin, citric acid, and enzymes.	Examples include production of ethanol, lactic acid, and biogas.

13. What is freeze drying and its advantages?

Freeze drying (lyophilization) is a preservation technique in which a biological material, such as microbial culture, is first frozen at low temperature and then the water is removed by sublimation under vacuum. This process converts the material into a dry form while maintaining the viability and stability of microorganisms.

Advantages:

- Maintains the viability and genetic characteristics of microbial cultures.
- Provides long-term storage of microorganisms.
- Prevents contamination and degradation of cultures.
- Dried cultures can be easily transported and stored.
- Microorganisms can be quickly revived by adding water or suitable medium.

14. Write names of chemical mutagens.

Chemical mutagens used for inducing mutations in microorganisms include:

- Ethyl methanesulfonate (EMS)
- Nitrosoguanidine (NTG or MNNG)
- Nitrous acid (HNO_2)
- Hydroxylamine
- Ethidium bromide
- Acridine dyes (acridine orange and acridine yellow)
- Base analogues such as 5-bromouracil

15. Name types of fermentation on the basis of stirring.

On the basis of stirring, fermentation is classified into the following types:

- ***Stirred fermentation:*** In this type, mixing of the fermentation medium is achieved by mechanical agitation using impellers or stirrers. It provides uniform distribution of nutrients, microorganisms, and oxygen throughout the medium and improves heat and mass transfer. It is commonly used in industrial fermenters.

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- ***Unstirred fermentation:*** In this type, no mechanical stirring is provided. Mixing occurs naturally due to gas movement, diffusion, or microbial activity. It is mainly used in anaerobic fermentation processes where oxygen supply and vigorous mixing are not required.

- ***Shaken fermentation:*** In this method, agitation is provided by shaking the culture vessels, such as flasks, on a shaker. It is mainly used for laboratory-scale studies and screening of microorganisms before large-scale fermentation.

16. On what factors fermentation can be classified?

Fermentation can be classified on the basis of the following factors:

- ***Type of microorganism used:*** Fermentation may be classified based on the organism involved, such as bacterial, fungal, yeast, or algal fermentation.
- ***Oxygen requirement:*** It can be classified as aerobic fermentation (requires oxygen) and anaerobic fermentation (occurs without oxygen).
- ***Mode of operation:*** Fermentation can be classified as batch, fed-batch, and continuous fermentation depending on the method of substrate addition and product removal.
- ***Type of product formed:*** It may be classified according to the end product, such as antibiotic fermentation, alcohol fermentation, enzyme fermentation, and organic acid fermentation.
- ***Agitation method:*** Fermentation can be classified as stirred, shaken, or unstirred fermentation based on the method of mixing.
- ***Scale of operation:*** It can be classified into laboratory, pilot-scale, and industrial-scale fermentation.

17. How microorganisms are stored under liquid nitrogen?

Microorganisms are stored under liquid nitrogen by cryopreservation at extremely low temperatures (about −196°C). The microbial cells are mixed with a protective agent such as glycerol or dimethyl sulfoxide (DMSO) to prevent ice crystal damage. The prepared culture is placed in sterile cryovials and stored in liquid nitrogen tanks. At this temperature, cellular metabolism is almost completely stopped, allowing microorganisms to remain viable and genetically stable for long periods. The cultures can be revived by rapid thawing and transferring them to a suitable growth medium.

18. What are biofertilizers?

Biofertilizers are preparations containing living microorganisms that improve soil fertility and enhance plant growth by increasing the availability of nutrients to plants. These microorganisms fix atmospheric nitrogen, solubilize phosphate, or produce growth-promoting substances. Examples of biofertilizers include Rhizobium (nitrogen fixation in leguminous plants), Azotobacter**,** Azospirillum**,** phosphate-solubilizing bacteria (PSB), and mycorrhizal fungi. They are eco-friendly and reduce the dependence on chemical fertilizers.

19. What is the role of pH in fermentation?

pH plays an important role in fermentation by affecting the growth, metabolism, and product formation of microorganisms. Each microorganism has an optimum pH range for maximum activity, and any change from this range can reduce enzyme activity and microbial growth. pH also influences nutrient availability, enzyme stability, oxygen transfer, and formation of unwanted by-products. Therefore, maintaining the optimum pH is essential for achieving high yield and quality of fermentation products.

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20. What is a bioweapon? Give some examples.

A bioweapon is a biological agent, such as a microorganism or toxin that is deliberately used to cause disease, death, or harm in humans, animals, or plants. These agents can include bacteria, viruses, fungi, or biological toxins. Examples of biological agents associated with bioweapon use include:

- ***Bacillus anthracis*** – causes anthrax.
- ***Yersinia pestis*** – causes plague.
- ***Variola virus*** – causes smallpox.
- ***Botulinum toxin produced by Clostridium botulinum*** – a potent biological toxin.

21. Name two methods to lower the dissolved oxygen.

Two methods to lower dissolved oxygen concentration are:

- ***Decrease aeration rate:*** Reducing the amount of air supplied through the sparger decreases the transfer of oxygen into the fermentation broth.
- ***Decrease agitation speed:*** Lowering the stirring speed reduces mixing and oxygen transfer efficiency, which results in a decrease in dissolved oxygen level.

22. Why mixing is done in fermentation?

Mixing is done in fermentation to ensure uniform distribution of microorganisms, nutrients, oxygen, and heat throughout the fermentation medium. It improves mass transfer and oxygen transfer by dispersing air bubbles and maintaining adequate dissolved oxygen levels. Mixing also prevents settling of cells and nutrients, maintains a uniform temperature and pH, and increases the efficiency of microbial growth and product formation.

23. How microorganisms are stored on agar plate?

Microorganisms are stored on agar plates by growing them on a suitable nutrient agar medium and maintaining the plates under controlled conditions. After sufficient growth, the agar plates are sealed with parafilm or stored in sterile containers and kept at low temperature (usually 4°C) to reduce microbial metabolism and prolong viability. Periodic subculturing is required to maintain the culture. This method is mainly used for short-term storage of microorganisms.

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Long Questions

1. Name five inhibitors used in fermentation.

Inhibitors Used in Fermentation

Inhibitors are substances added to fermentation media to suppress the growth of unwanted microorganisms, control contamination, or selectively inhibit specific metabolic activities. Some commonly used inhibitors are:

- ***Penicillin:*** Penicillin is an antibiotic that inhibits the growth of susceptible bacteria by interfering with cell wall synthesis. It is used for selective inhibition of bacterial contaminants and in mutation studies.
- ***Sodium azide:*** Sodium azide acts as a metabolic inhibitor by blocking respiratory enzymes, particularly cytochrome oxidase. It is used to inhibit unwanted microbial growth during selection procedures.
- ***Chloroform:*** Chloroform is a chemical antimicrobial agent that disrupts cell membranes and inhibits the growth of many microorganisms. It is used for controlling contamination.
- ***Phenol:*** Phenol is a disinfectant and antimicrobial compound that denatures proteins and damages microbial cell membranes, resulting in inhibition of microbial growth.
- ***Sodium benzoate:*** Sodium benzoate is a preservative used to inhibit the growth of unwanted bacteria and fungi, especially in acidic fermentation environments.

2. Explain and write stoichiometric equation.

Stoichiometric Equation in Fermentation

A stoichiometric equation represents the quantitative relationship between substrates consumed and products formed during a fermentation process. It is based on the principle of mass balance, where the amount of elements entering the fermentation system equals the amount leaving as products, biomass, and by-products. The general stoichiometric equation for microbial growth is:

$$C_6H_{12}O_6 + aO_2 + bNH_3 \rightarrow cC_xH_yO_z + dCO_2 + eH_2O$$

Where:

- $C_6H_{12}O_6$ = Carbon source (glucose)
- O_2 = Oxygen supplied during aerobic fermentation
- NH_3 = Nitrogen source
- $C_xH_\gamma O_z$ = Microbial biomass
- CO_2 = Carbon dioxide produced
- H_2O = Water formed

For alcoholic fermentation by yeast:

$$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 + Energy$$

Glucose is converted into ethanol, carbon dioxide, and energy by microorganisms under anaerobic conditions. Stoichiometric equations are useful for calculating substrate utilization, biomass formation, oxygen requirement, and product yield during fermentation.

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3. Explain process of random mutagenesis.

Random Mutagenesis

Random mutagenesis is a technique used to produce genetic variations in microorganisms by exposing them to physical or chemical mutagens without targeting a specific gene. It is widely used in industrial microbiology to obtain improved microbial strains with higher productivity, better tolerance, or desirable characteristics.

Process of Random Mutagenesis:

- ***Selection of microorganism:*** A suitable microbial strain with desired industrial properties is selected as the starting organism.
- ***Exposure to mutagen:*** The microbial cells are treated with mutagenic agents such as UV radiation, X-rays, ethyl methanesulfonate (EMS), nitrosoguanidine (NTG), or nitrous acid. These agents cause random changes in the DNA sequence.
- ***Mutation induction:*** The mutagens produce different types of genetic alterations, including base substitutions, deletions, or insertions, resulting in mutant microorganisms.
- ***Screening and selection:*** The treated cells are grown on suitable media, and mutants with improved properties are identified through screening methods.
- ***Isolation and evaluation of mutants:*** Promising mutants are isolated, tested for stability, and evaluated for improved production of desired fermentation products.
- ***Industrial application:*** Selected improved strains are used in large-scale fermentation processes for the production of antibiotics, enzymes, amino acids, and other industrial products.

4. What are different phenotypic techniques to identify microorganisms?

Phenotypic Techniques for Identification of Microorganisms

Phenotypic identification techniques are methods used to identify microorganisms based on their observable characteristics, including morphology, biochemical activities, and physiological properties.

1. Morphological Characteristics

Microorganisms are identified by observing their cell shape, size, arrangement, and staining characteristics. Methods such as Gram staining, acid-fast staining, and microscopic examination are used. Colony characteristics such as color, shape, size, texture, and pigmentation on culture media are also studied.

2. Biochemical Tests

These tests identify microorganisms based on their metabolic activities and enzyme production. Common tests include catalase test, oxidase test, urease test, indole test, and carbohydrate fermentation tests**.** The biochemical profile of an organism helps in species identification.

3. Physiological Tests

Identification is based on the ability of microorganisms to grow under different environmental conditions. Factors studied include temperature range, pH tolerance, salt concentration, and oxygen requirement.

4. Nutritional Tests

Microorganisms are identified by their ability to utilize different carbon and nitrogen sources. Growth patterns on specific nutrient media help differentiate between species.

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5. Serological Methods

These methods identify microorganisms using antigen–antibody reactions. Techniques such as agglutination tests and ELISA are used for detection and identification.

6. Antibiotic Sensitivity Testing

Microorganisms are identified based on their response to different antibiotics. The pattern of resistance or sensitivity can help differentiate closely related organisms.

7. Automated Phenotypic Identification

Instruments such as VITEK and Biolog systems identify microorganisms by analyzing multiple biochemical and physiological characteristics simultaneously.

5. What is streak plate method?

Streak Plate Method

The streak plate method is a microbiological technique used for the isolation and separation of pure cultures of microorganisms from a mixed population. In this method, a small amount of microbial culture is spread over the surface of a solid agar medium using a sterile inoculating loop. The streaking process gradually reduces the number of cells deposited on the agar surface, allowing individual cells to form separate colonies.

Procedure:

1. A sterile agar plate containing suitable growth medium is prepared.
2. A small amount of microbial culture is taken using a sterile inoculating loop.
3. The culture is streaked over the agar surface in a specific pattern to dilute the cells gradually.
4. The plate is incubated under suitable conditions for microbial growth.
5. Individual colonies develop from single microbial cells and can be picked for obtaining pure cultures.

Types of streaking patterns:

- Quadrant streak method
- T-streak method
- Continuous streak method

Applications:

- Isolation of pure microbial cultures.
- Separation of different microorganisms from mixed samples.
- Identification and study of microbial characteristics.
- Maintenance of laboratory cultures.

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6. What factors affect the growth of microbes?

Factors Affecting the Growth of Microbes

Microbial growth is influenced by various environmental and nutritional factors that affect the rate of cell division, metabolism, and product formation.

- ***Nutrient Availability:*** Microorganisms require essential nutrients such as carbon, nitrogen, minerals, vitamins, and growth factors for growth. The type and concentration of nutrients in the medium directly affect microbial growth and productivity.
- ***Temperature:*** Temperature affects enzyme activity and metabolic reactions of microorganisms. Each microorganism has an optimum temperature for growth. Extreme temperatures can inhibit growth or cause cell death.
- ***pH:*** pH influences enzyme activity, nutrient availability, and cellular processes. Most microorganisms grow best within a specific pH range; changes beyond this range can reduce growth.
- ***Oxygen Availability:*** Oxygen concentration affects microorganisms based on their oxygen requirements. Aerobic microorganisms require oxygen for growth, while anaerobic microorganisms grow in the absence of oxygen.
- ***Moisture and Water Activity:*** Water is essential for microbial metabolism and nutrient transport. Low water availability can inhibit microbial growth, while adequate moisture promotes growth.
- ***Osmotic Pressure and Salt Concentration:*** High concentrations of salts or sugars can create osmotic stress, causing water loss from cells and reducing microbial growth.
- ***Agitation and Mixing:*** Proper agitation ensures uniform distribution of nutrients, oxygen, and heat in the fermentation medium. Poor mixing can limit microbial growth.
- ***Inhibitory Substances:*** Toxic compounds, antibiotics, metabolic waste products, and other inhibitors can reduce or stop microbial growth.
- ***Pressure:*** Pressure conditions, including gas pressure and osmotic pressure, can influence microbial growth and metabolic activity.
- ***Time of Incubation:*** Microorganisms pass through different growth phases (lag, log, stationary, and death phases). The duration of incubation affects cell number and product formation.

7. Write a note on agitation and aeration.

Agitation

Agitation is the process of mixing the fermentation medium mechanically using impellers or stirrers. It ensures uniform distribution of microorganisms, nutrients, oxygen, and heat throughout the fermenter. Agitation improves mass transfer, prevents settling of cells, maintains a uniform temperature and pH, and increases the contact between microbial cells and dissolved oxygen.

Functions of agitation:

- Maintains homogeneous mixing of fermentation broth.
- Improves oxygen transfer from air bubbles to the liquid phase.
- Enhances heat transfer and prevents temperature gradients.
- Keeps microbial cells suspended in the medium.
- Improves nutrient availability and product formation.

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Factors affecting agitation:

- Type, size, and speed of impeller.
- Viscosity of fermentation medium.
- Volume of culture broth.
- Presence of antifoaming agents.

Aeration

Aeration is the process of supplying sterile air or oxygen to the fermentation medium to provide oxygen required by aerobic microorganisms. Air is introduced into the fermenter through a sparger, which produces small bubbles and increases oxygen transfer into the broth.

Functions of aeration:

- Provides oxygen for microbial respiration and growth.
- Maintains dissolved oxygen concentration in the fermentation broth.
- Removes carbon dioxide produced during metabolism.
- Supports high microbial activity and product yield.

Factors affecting aeration:

- Air flow rate.
- Bubble size.
- Oxygen solubility.
- Agitation speed.
- Temperature and pressure of the system.

8. Write criteria for process organism by Bull *et al.* (1979).

Criteria for Process Organism by Bull et al. (1979)

According to Bull *et al.* (1979)**,** an industrial microorganism used in a fermentation process should possess the following characteristics:

- ***High productivity:*** The organism should produce a large amount of desired product in a short period of time with high yield.
- ***Genetic stability:*** The organism should maintain its desirable characteristics and productivity during repeated subculturing and large-scale fermentation.
- ***Non-pathogenic nature:*** The microorganism should be safe to handle and should not produce harmful toxins or cause diseases.
- ***Ability to utilize inexpensive substrates:*** The organism should efficiently grow on low-cost raw materials such as agricultural wastes and industrial by-products.
- ***Rapid growth rate:*** The organism should grow quickly and produce sufficient biomass or product within a short fermentation period.
- ***Tolerance to environmental conditions:*** The organism should withstand changes in pH, temperature, osmotic pressure, and high concentrations of substrates or products.
- ***Ease of cultivation and maintenance:*** The organism should be easy to grow, preserve, and maintain under laboratory and industrial conditions.
- ***Efficient product recovery:*** The desired product should be easily separated and purified from the fermentation broth.

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- ***Minimal production of unwanted by-products:*** The organism should produce fewer undesirable metabolites to simplify downstream processing.
- ***Ability to grow in large-scale fermenters:*** The organism should perform well under industrial fermentation conditions with proper aeration, agitation, and nutrient supply.

9. Explain dehydrated form of organisms.

Dehydrated Form of Organisms

Dehydration is a method of preserving microorganisms by removing water from microbial cells, which reduces metabolic activity and prevents deterioration. In this state, microorganisms remain inactive but viable and can be revived when suitable moisture and nutrients are provided.

Methods of Dehydration:

- ***Drying on inert carriers:*** Microorganisms are dried on materials such as soil, silica gel, or filter paper. The reduced moisture content helps maintain the viability of cultures for storage.
- ***Freeze drying (Lyophilization):*** In this method, microbial cells are first frozen and then water is removed by sublimation under vacuum. It is one of the most effective methods for long-term preservation of microorganisms.
- ***Air drying:*** Microbial cultures are dried using controlled airflow. This method is simple but may reduce the survival rate of sensitive microorganisms.
- ***Spray drying:*** A microbial suspension is sprayed into a stream of hot air, causing rapid evaporation of water and formation of dried microbial powder.

Advantages of Dehydrated Forms:

- Provides long-term storage of microbial cultures.
- Reduces contamination risk.
- Maintains genetic stability of microorganisms.
- Requires less storage space compared to liquid cultures.
- Allows easy transport and distribution of microbial strains.

Limitations:

- Some microorganisms may lose viability during dehydration.
- Sensitive cells may require protective agents such as glycerol, skim milk, or sugars during the drying process.

10. Explain aerobic and anaerobic fermentation.

Aerobic Fermentation

Aerobic fermentation is a type of fermentation process in which microorganisms grow and produce desired products in the presence of oxygen. Oxygen is supplied to the fermentation medium through aeration and agitation to support microbial respiration and metabolism. Aerobic microorganisms use oxygen as the final electron acceptor, resulting in efficient energy production and high biomass formation.

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Characteristics of aerobic fermentation:

- Requires continuous supply of oxygen.
- Uses aerobic microorganisms such as bacteria and fungi.
- Requires proper aeration and agitation for oxygen transfer.
- Produces more energy due to complete oxidation of substrates.
- Commonly used for production of antibiotics, enzymes, amino acids, and organic acids.

Examples:

- Production of penicillin by *Penicillium chrysogenum*.
- Production of citric acid by *Aspergillus niger*.
- Production of streptomycin by *Streptomyces griseus*.

Anaerobic Fermentation

Anaerobic fermentation is a fermentation process in which microorganisms grow and produce products in the absence of oxygen. These microorganisms obtain energy by breaking down organic substrates through anaerobic metabolic pathways.

Characteristics of anaerobic fermentation:

- Does not require oxygen supply.
- Carried out by anaerobic microorganisms.
- Produces less energy compared to aerobic fermentation.
- Requires strict control to prevent oxygen entry.
- Produces reduced end products such as alcohols, acids, and gases.

Examples:

- Production of ethanol by *Saccharomyces cerevisiae*.
- Production of lactic acid by *Lactobacillus* species.
- Production of biogas by methanogenic microorganisms.

11. Name five compounds that mimic the ideal antifoam agents.

Compounds That Mimic Ideal Antifoam Agents

An ideal antifoam agent should rapidly destroy foam, prevent further foam formation, be non-toxic to microorganisms, and not interfere with oxygen transfer or product formation. Some compounds that possess most of these properties are:

- ***Silicone oils (Polydimethylsiloxane):*** Silicone oils are highly effective antifoams used in fermentation industries. They reduce surface tension and quickly collapse foam bubbles. They are chemically stable and non-toxic to most microorganisms.
- ***Polypropylene glycol (PPG):*** PPG is a synthetic antifoaming agent widely used in fermenters. It prevents foam formation by reducing bubble stability. It has good compatibility with fermentation systems.
- ***Vegetable oils:*** Oils such as soybean oil, cottonseed oil, and olive oil act as natural antifoams. They reduce surface tension and control excessive foam formation.
- ***Mineral oils:*** Mineral oils are hydrocarbon-based antifoams that spread over the foam surface and cause bubble collapse. They are stable and effective at low concentrations.
- ***Fatty acids and fatty acid esters:*** Compounds such as oleic acid and stearates reduce foam stability by altering surface properties. They are commonly used in microbial fermentation processes.

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12. Write properties of culture that are used in fermentation.

Properties of Microbial Culture Used in Fermentation

A microorganism selected for industrial fermentation should possess desirable characteristics to ensure efficient production, high yield, and easy handling. Important properties include:

- ***High productivity:*** The culture should produce a large amount of desired product in a short fermentation period. It should efficiently convert substrates into valuable products.
- ***Genetic stability:*** The microorganism should maintain its desirable genetic characteristics during repeated subculturing and large-scale production. It should not easily lose its ability to produce the required product.
- ***Rapid growth rate:*** The culture should grow quickly and produce sufficient biomass. Rapid growth reduces fermentation time and increases productivity.
- ***Non-pathogenic nature:*** The organism should be safe for workers and the environment. It should not produce harmful toxins or pathogenic compounds.
- ***Ability to utilize inexpensive substrates:*** The culture should grow on low-cost raw materials such as agricultural wastes, molasses, and starch-based materials.
- ***Tolerance to environmental conditions:*** The organism should tolerate changes in pH, temperature, osmotic pressure, and high concentrations of nutrients or products.
- ***High yield and product specificity:*** The culture should produce maximum amounts of the desired product with minimum formation of unwanted by-products.
- ***Easy recovery of product:*** The microorganism should produce products that can be easily separated and purified from the fermentation broth.
- ***Resistance to contamination:*** The culture should be able to compete with or tolerate minor contamination during fermentation.
- ***Suitable morphological characteristics:*** The organism should have a form that allows good mixing, oxygen transfer, and easy harvesting in industrial fermenters.

13. What are inorganic sources of nitrogen in fermentation?

Inorganic Sources of Nitrogen in Fermentation

Nitrogen is an essential nutrient required by microorganisms for the synthesis of proteins, nucleic acids, enzymes, and other cellular components. In fermentation media, inorganic nitrogen sources are commonly used because they are inexpensive and readily available. Important inorganic nitrogen sources include:

- ***Ammonium salts:*** Ammonium salts are widely used as nitrogen sources because they are easily utilized by microorganisms. Examples include ammonium sulfate ($(NH_4)_2SO_4$), ammonium chloride (NH_4Cl), and ammonium nitrate (NH_4NO_3).
- ***Ammonia (NH_3):*** Ammonia is a simple and readily available nitrogen source. It is also used for controlling the pH of fermentation media by neutralizing acids produced during microbial growth.
- ***Nitrates:*** Nitrate salts provide nitrogen in oxidized form and are utilized by some microorganisms after reduction. Example: sodium nitrate ($NaNO_3$) and potassium nitrate (KNO_3).
- ***Urea:*** Urea is an inorganic nitrogen source that is hydrolyzed by microorganisms to release ammonia. It is commonly used due to its high nitrogen content and low cost.
- ***Ammonium phosphate:*** Compounds such as diammonium phosphate (DAP) and monoammonium phosphate (MAP) provide both nitrogen and phosphorus required for microbial growth.

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14. What are some industrially important enzymes and their inducers?

Industrially Important Enzymes and Their Inducers

Inducers are substances that stimulate microorganisms to produce specific enzymes by activating enzyme synthesis. Many industrial enzymes are produced through fermentation using suitable microorganisms and inducers.

Enzyme	Microorganism	Inducer
Amylase	*Bacillus subtilis*, *Aspergillus oryzae*	Starch, maltose
Protease	*Bacillus licheniformis*, *Aspergillus* species	Proteins, casein, peptones
Cellulase	*Trichoderma reesei*	Cellulose, carboxymethyl cellulose (CMC)
Pectinase	*Aspergillus niger*	Pectin, plant extracts
Lactase (β-galactosidase)	*Kluyveromyces lactis*	Lactose
Lipase	*Candida rugosa*, *Aspergillus niger*	Oils, fatty acids, triglycerides
Invertase	*Saccharomyces cerevisiae*	Sucrose
Chitinase	*Bacillus* species, fungi	Chitin

15. What factors affect the rate of diffusion of oxygen through the boundary layer?

Factors Affecting the Rate of Diffusion of Oxygen Through the Boundary Layer

The diffusion of oxygen through the boundary layer surrounding microbial cells is an important step in oxygen transfer during fermentation. The rate of oxygen diffusion depends on several physical and biological factors:

1. Thickness of Boundary Layer

The boundary layer is a thin layer of liquid surrounding microbial cells and air bubbles where oxygen transfer occurs. A thinner boundary layer increases the rate of oxygen diffusion, while a thicker layer decreases oxygen transfer.

2. Oxygen Concentration Gradient

Oxygen diffusion depends on the difference between oxygen concentration at the surface of the liquid and inside the microbial cell. A higher concentration gradient increases the driving force for oxygen transfer.

3. Agitation Rate

Increased agitation reduces the thickness of the boundary layer and improves mixing. It increases contact between oxygen bubbles and the fermentation broth, enhancing oxygen diffusion.

4. Aeration Rate

Higher aeration provides more oxygen bubbles in the medium, increasing oxygen availability. It improves the oxygen transfer rate by increasing the gas–liquid contact area.

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5. Temperature

Temperature affects oxygen solubility in the fermentation broth. Higher temperatures generally decrease oxygen solubility, reducing oxygen transfer efficiency.

6. Viscosity of Fermentation Medium

Highly viscous media slow down oxygen diffusion because movement of oxygen molecules becomes difficult. Low-viscosity media allow faster oxygen transfer.

7. Bubble Size

Smaller air bubbles provide a larger surface area for oxygen transfer and increase diffusion rate. Large bubbles have less surface area and lower oxygen transfer efficiency.

8. Cell Concentration

High microbial cell density increases oxygen demand and may reduce available dissolved oxygen. Excessive biomass can increase medium viscosity and limit oxygen diffusion.

9. Pressure

Increased pressure increases oxygen solubility in the liquid, improving oxygen transfer. Reduced pressure decreases oxygen availability.

10. Composition of Fermentation Medium

Components such as proteins, salts, and oils can affect oxygen solubility and diffusion properties of the medium.

16. Explain methods of antibiotic screening.

Methods of Antibiotic Screening

Antibiotic screening is a process used to identify and evaluate microorganisms that produce antimicrobial compounds. It involves detecting microorganisms capable of inhibiting the growth of other microbes and selecting promising antibiotic-producing strains for further study.

1. Primary Screening

Primary screening is the initial detection method used to identify microorganisms with antibiotic-producing ability. It is a qualitative method that determines whether a microorganism produces an antimicrobial substance.

Methods of primary screening include:

a) Crowded plate technique: A large number of microorganisms are grown on an agar plate. Colonies producing antibiotics inhibit the growth of surrounding microorganisms, forming clear zones. Colonies showing inhibition are selected for further testing.

b) Cross-streak method: The antibiotic-producing microorganism is streaked on agar medium. Test organisms are streaked perpendicular to it. Inhibition of growth near the producer organism indicates antibiotic production.

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c) Agar diffusion method: The culture filtrate or antibiotic extract is placed on agar containing a test microorganism. A clear zone of inhibition around the sample indicates antimicrobial activity.

2. Secondary Screening

Secondary screening is a quantitative evaluation performed after primary screening to confirm antibiotic production and measure its effectiveness.

Methods of secondary screening include:

a) Fermentation studies: Selected microorganisms are grown in liquid fermentation media. Antibiotic production is measured under controlled conditions.

b) Bioassay: The potency of an antibiotic is determined by measuring the inhibition of sensitive test organisms. It helps compare the activity of produced antibiotics with standard antibiotics.

c) Chemical analysis: Chromatographic and spectroscopic techniques are used to identify and quantify the antibiotic compound.

3. Selection and Improvement of Strains

- High-producing strains are selected after screening.
- Mutagenesis and genetic improvement techniques may be used to increase antibiotic yield.

17. Write advantages of fats and oils as a source of carbon in fermentation.

Advantages of Fats and Oils as a Source of Carbon in Fermentation

Fats and oils are used as carbon sources in fermentation media because they provide a concentrated source of energy and carbon required for microbial growth and product formation. They are commonly used in the production of antibiotics, organic acids, and other industrial products.

Advantages:

- ***High carbon and energy content:*** Fats and oils contain more carbon atoms and provide higher energy compared to carbohydrates. They support vigorous microbial growth and increase product yield.
- ***Slow and continuous carbon supply:*** Fats and oils are gradually utilized by microorganisms, providing a continuous supply of carbon during fermentation. This helps prevent excessive growth and unwanted metabolic products.
- ***Increase in product formation:*** Certain microorganisms produce higher amounts of secondary metabolites, such as antibiotics and enzymes, when grown on oil-based carbon sources.
- ***Low osmotic pressure:*** Oils do not significantly increase the osmotic pressure of the fermentation medium compared with high concentrations of sugars. This allows better microbial growth and metabolism.
- ***Reduced foam formation:*** Some oils act as natural antifoaming agents by decreasing surface tension and reducing excessive foam production.
- ***Better oxygen transfer:*** Oils can improve oxygen availability in some fermentation systems by increasing oxygen solubility in the medium.
- ***Availability and low cost:*** Many fats and oils are easily available from plant and animal sources and can be economical substrates for industrial fermentation.
- ***Utilization of waste materials:*** Industrial waste oils and agricultural by-products can be used as carbon sources, reducing production costs and waste disposal problems.

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18. Write down five major groups of commercially important fermentations.

Major Groups of Commercially Important Fermentations

Commercial fermentation processes are classified based on the type of products produced. The major groups of industrially important fermentations are:

1. Production of Microbial Cells (Biomass)

In this type of fermentation, microorganisms themselves are the final product. The process involves large-scale cultivation of microbial cells for use as food, feed, or inoculants. Examples include baker's yeast production, single-cell protein (SCP), and biofertilizer production.

2. Production of Microbial Metabolites

Microbial metabolites are compounds produced during microbial growth and metabolism.

a) Primary metabolites

- Produced during active growth phase.
- Examples include amino acids (glutamic acid, lysine), organic acids (citric acid, lactic acid), and ethanol.

b) Secondary metabolites

- Produced mainly during the stationary phase.
- Examples include antibiotics (penicillin, streptomycin) and pigments.

3. Production of Industrial Enzymes

- Microorganisms are cultivated to produce commercially valuable enzymes.
- These enzymes are used in food, pharmaceutical, textile, and detergent industries.
- Examples include amylase, protease, cellulase, lipase, and pectinase.

4. Transformation of Substrates (Bioconversion)

- In this process, microorganisms convert one compound into another valuable product through enzymatic reactions.
- It is used for producing specific chemicals and pharmaceutical compounds.
- Examples include steroid transformation and production of modified antibiotics.

5. Production of Recombinant Products

- Genetically modified microorganisms are used to produce valuable biological products.
- These products are formed by introducing foreign genes into microbial cells.
- Examples include insulin, vaccines, hormones, and therapeutic proteins.

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19. Write down the names of five compounds that meet most of the ideal antifoam requirements.

Compounds That Meet Most of the Ideal Antifoam Requirements

An ideal antifoam agent should have rapid foam-breaking ability, low toxicity, chemical stability, low cost, and should not interfere with microbial growth or oxygen transfer. Compounds that meet most of these requirements include:

- ***Polydimethylsiloxane (Silicone oil):*** It is one of the most widely used antifoaming agents in fermentation. It rapidly reduces foam formation and is effective at low concentrations. It is stable and has low toxicity toward microorganisms.
- ***Polypropylene glycol (PPG):*** PPG is a synthetic antifoam agent used in industrial fermenters. It reduces surface tension and prevents the formation of stable foam.
- ***Vegetable oils:*** Oils such as soybean oil, olive oil, and cottonseed oil act as natural antifoaming agents. They are biodegradable and reduce foam by spreading over the foam surface.
- ***Mineral oils:*** Mineral oils are hydrocarbon-based antifoams that collapse foam bubbles by disrupting the liquid film around them. They are chemically stable and effective in fermentation processes.
- ***Fatty acid esters:*** Compounds such as sorbitan esters (Spans) and fatty acid derivatives reduce foam stability. They are useful because they have good compatibility with fermentation media.

20. How to provide aeration at lab, pilot, and industrial scale?

Aeration is the process of supplying sterile air or oxygen to a fermentation system to maintain adequate dissolved oxygen levels required for the growth and activity of aerobic microorganisms. The method of aeration depends on the scale of fermentation.

1. Laboratory Scale Aeration

At laboratory scale, aeration is usually provided by shaking flasks on rotary or orbital shakers. The movement of the flask increases contact between the culture medium and air, allowing oxygen transfer. In laboratory fermenters, sterile air is supplied through spargers, and agitation is provided by small impellers to improve oxygen transfer. Parameters such as agitation speed, air flow rate, and dissolved oxygen are controlled for microbial growth studies.

2. Pilot Scale Aeration

Pilot-scale fermenters use systems similar to industrial fermenters but at a smaller volume. Sterile air is introduced through spargers located at the bottom of the fermenter. Mechanical agitation using impellers disperses air bubbles throughout the medium. Air flow rate and agitation speed are adjusted to maintain suitable dissolved oxygen concentration. Pilot studies help optimize aeration conditions before industrial production.

3. Industrial Scale Aeration

In industrial fermenters, large quantities of sterile air are supplied continuously through spargers. Different types of spargers such as ring spargers, nozzle spargers, and porous spargers are used to produce fine air bubbles. Large impellers provide efficient mixing and increase oxygen transfer from air bubbles to the fermentation broth. Aeration is controlled using dissolved oxygen sensors and automatic control systems. Pure oxygen may be added when oxygen demand is very high and normal air supply is insufficient.

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